


# DM481e

## User Manual

*Version 2.1, 07-2017*

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## A. General Information

- Refer to the User Manual prepared by the manufacturer for proper usage of the instrument.
- If the product is used in a manner not specified, the protection provided by the instrument warranty may be impaired.
- Return the instrument to Aemulus for service and repair to ensure that safety features are maintained.

**WARNING**

**The WARNING heading explains hazards that might occur if an operating procedure, practice, or the like that, if not correctly performed or adhered to might result in personal injury or death. Always read the associated information very carefully and make sure the indicated information is met before performing the indicated procedure.**

**CAUTION**

**The CAUTION heading in the user documentation explains hazards that could damage the instrument. Such damage may invalidate the warranty.**

## B. Identification and Discouragement of Hazardous Applications

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**CAUTION**

*Main Power*

- Ensure that the mains power is turned off before plugging or removing the PXI module into or from the chassis.
- Make sure the power cord is plugged properly to the chassis before applying power.
- The chassis that carries the instrument has to be plugged into a grounded (earthed) power outlet. By not doing so, it will defeat the power cord safety ground feature.
- Ensure that the power is removed from the product before connecting/disconnecting the cables to the product connectors.

---

**WARNING**

*Instrument Cables / Connectors*

- Inspection of the connecting cable and test leads for possible wear, cracks, or breaks must be done before each use.
- Make sure the cables used for the product are rated to the specified voltage and current level.

---

**WARNING**

*Instrument Output*

- Extreme caution is to be practiced when a shock hazard is present. Lethal voltage may be present on cable connector jacks.
- When power is applied to the circuit under test, do not touch the product and test cables or test leads.
- Do not touch the connector jacks or cables when the output of the product is turned on.

---

**WARNING**

*Hot Surface*

- If the DM481e has been in use, it might exceed safe handling temperatures and cause burns. Allow some cooling time before removing the DM481e from the chassis.

**WARNING**

- The fan speed of the chassis that carries the instrument has to be in HIGH speed to allow proper heat ventilation.

*Operating Temperature*

## C. Identification of Authorized Personnel

| | |
|---|---|
| *End user* | Individual or group who is responsible for the use and maintenance of the product, as well as being responsible for ensuring that the product is operated within its specifications. Knowledge on electrical safety procedures and proper use of the product are prerequisite before operating the product. |
| *Maintenance personnel* | Individual or group who is responsible for performing routine procedures on the product to keep it operating properly. Any service should only be done by qualified service personnel. |
| *Service personnel* | Installation and service procedures may only be carried out by properly trained service personnel. Knowledge on working with live circuits, performing safe installations, and repairing the products are prerequisite for this individual or group. |

## Section 1: List of Abbreviations and Nomenclature

| Abbreviation | Description |
|---|---|
| VHDCI | Very High Density Cable Interconnect |
| FH | Force-High |
| SH | Sense-High |
| DIO | Digital Input/Output |
| ESD | Electrostatic Discharge |
| DM | Digital Module |
| NPLC | Number of Power Line Cycle |
| DUT | Device Under Test |
| $DDR_3$ | Third Generation Double Data Rate |
| SPI | Serial Peripheral Interface |
| PMU | Pin Measurement Unit |
| PE | Pin Electronics |
| PLL | Phase-Locked Loop |
| DCL | DC Active Load |
| ATE | Automated test equipment |
| DICV | Drive Current, Clamp Voltage |
| DVCI | Drive Voltage, Clamp Current |
| PCIe | Peripheral Component Interconnect Express |
| PXI | PC-based Platform Extensions for Instrumentation |
| PPMU | Per Pin Parametric Measurement Unit |
| NC | No Connection |
| GND | Ground |
| ACT | Active |
| PWR | Power |
| LED | Light-Emitting Diode |
| DPIN | Digital Pin |
| SMU | Source Measure Unit |
| Tset | Timing Sets |

## Section 2: Quick Start

### 2.1 Product Start-up Configuration

Each product is carefully inspected before it leaves the factory. Upon receipt and before unpacking the product, please inspect the unit for any obvious damage that may occur during shipment. If any damage is found, notify the instrument manufacturer immediately.

Please verify that the following items are included in the original packing of this product. Contact the instrument manufacturer immediately if any of these items are found missing.

- Reference / Drivers CD-ROM

The product is designed to be used in an indoor environment, an area with low condensation and a temperature of between 23°C ± 10°C. Ensure that the area where the product is installed is free from the presence of any explosive or flammable fumes or gases.

**CAUTION**

The product is shipped in materials that prevent static damage to the module. The module should only be removed from the packaging in an anti-static area ensuring that correct anti-static precautions are taken. Store all modules in anti-static envelopes when not in use.

## 2.2 Product Software & Hardware Installation

Refer to installation guide.

## Section 3: Operation

### 3.1 Front Panel Connectors

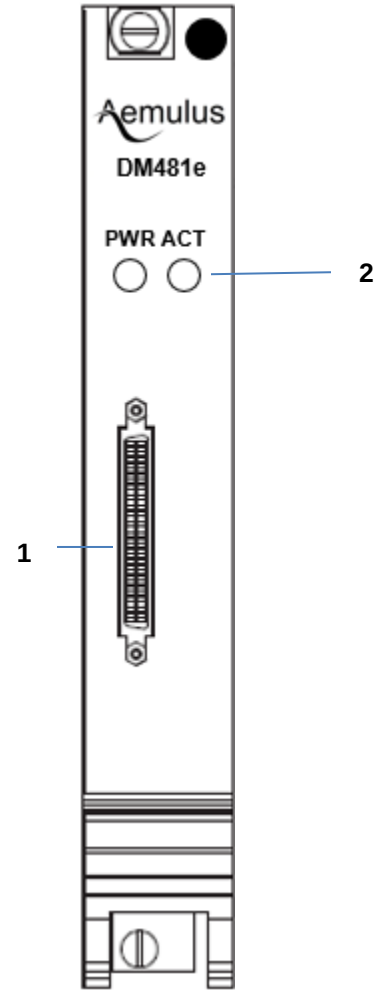



Figure 1: DM481e Front Panel

1. OUTPUT (VHDCI, 68-Position, Receptacle)

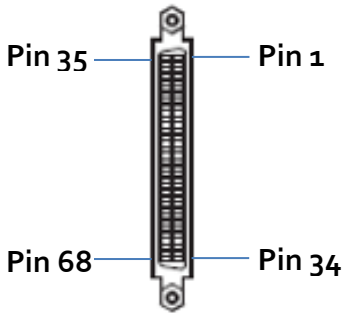



Figure 2: Pin Orientation

2. LED indicators

| Symbol | Indicator | Description |
|---|---|---|
| ACT | Active | **Green** when there is activity on PCIe links.<br>**Red** when factory image is being loaded. |
| PWR | Power | **Green** when it is powered up successfully.<br>**Red** when<br>▪ On-board PLL is not locked.<br>▪ DDR3 memory module is not detected or error occurs.<br>▪ Operating temperature is exceeded.<br>▪ Power supply failure. |

Table 1: Front Panel LED Indicators

## Pin Assignment:

| Pin | Signal Name (Primary) | Description |
|---|---|---|
| 1 | NC | - |
| 2 | NC | - |
| 3 | NC | - |
| 4 | NC | - |
| 5 | NC | - |
| 6 | NC | - |
| 7 | CH11_SH | Pin 11 Sense-High |
| 8 | GND | Ground |
| 9 | CH11_FH | Pin 11 Force-High |
| 10 | GND | Ground |
| 11 | CH9_FH | Pin 9 Force-High |
| 12 | NC | - |
| 13 | CH7_SH | Pin 7 Sense-High |
| 14 | GND | Ground |
| 15 | CH7_FH | Pin 7 Force-High |
| 16 | NC | - |
| 17 | CH9_SH | Pin 9_Sense-High |
| 18 | NC | - |
| 19 | NC | - |
| 20 | NC | - |
| 21 | EXT_TRIGIN0 | External Trigger Input 0 |
| 22 | GND | Ground |
| 23 | EXT_TRIGOUT0 | External Trigger Output 0 |
| 24 | GND | Ground |
| 25 | CH5_SH | Pin 5 Sense-High |
| 26 | GND | Ground |
| 27 | CH5_FH | Pin 5 Force-High |
| 28 | GND | Ground |
| 29 | CH3_FH | Pin 3 Force-High |
| 30 | CH3_SH | Pin 3 Sense-High |
| 31 | CH1_FH | Pin 1 Force-High |
| 32 | GND | Ground |
| 33 | CH1_SH | Pin 1 Sense-High |
| 34 | NC | - |

Table 2: I/O Pin-out (Pins 1-34)

| Pin | Signal Name (Primary) | Description |
|---|---|---|
| 35 | NC | - |
| 36 | NC | - |
| 37 | NC | - |
| 38 | NC | - |
| 39 | CH8_SH | Pin 8_Sense-High |
| 40 | NC | - |
| 41 | CH10_SH | Pin 10 Sense-High |
| 42 | GND | Ground |
| 43 | CH10_FH | Pin 10 Force-High |
| 44 | GND | Ground |
| 45 | CH8_FH | Pin 8 Force-High |
| 46 | NC | - |
| 47 | CH6_SH | Pin 6 Sense-High |
| 48 | GND | Ground |
| 49 | CH6_FH | Pin 6 Force-High |
| 50 | GND | Ground |
| 51 | EXT_TRIGOUT1 | External Trigger Output 1 |
| 52 | GND | Ground |
| 53 | EXT_TRIGIN1 | External Trigger Input 1 |
| 54 | NC | - |
| 55 | NC | - |
| 56 | NC | - |
| 57 | COMMON_SL | Common Sense Low |
| 58 | NC | - |
| 59 | CH4_SH | Pin 4 Sense-High |
| 60 | GND | Ground |
| 61 | CH4_FH | Pin 4 Force-High |
| 62 | GND | Ground |
| 63 | CH2_FH | Pin 2 Force-High |
| 64 | CH2_SH | Pin 2 Sense-High |
| 65 | CH0_FH | Pin 0 Force-High |
| 66 | GND | Ground |
| 67 | CH0_SH | Pin 0 Sense-High |
| 68 | NC | - |

Table 3: I/O Pin-out (Pins 35-68)

| Signal Name | Description |
|---|---|
| PINx_FH | The I/O for a pin for all operating modes (PMU, VECTOR and DIO). Refer to programming manual for details. |
| PINx_SH | The remote sense line when operating in PMU mode. |
| COMMON_SL | The common sense low line. |
| EXT_TRIGOUTx | External trigger output from the module. |
| EXT_TRIGINx | External trigger input to the module. |

Table 4: Pin Function Description

## 3.2 Soft Front Panel

The soft front panel provides a graphical interface for operating the module. It is intended for testing and diagnosing, for demonstration and training, and for basic operation of the module. It represents most of the functions available in the instrument driver. It is not however a comprehensive application suitable for measurements; for this, remote programming the module.

### General Functions

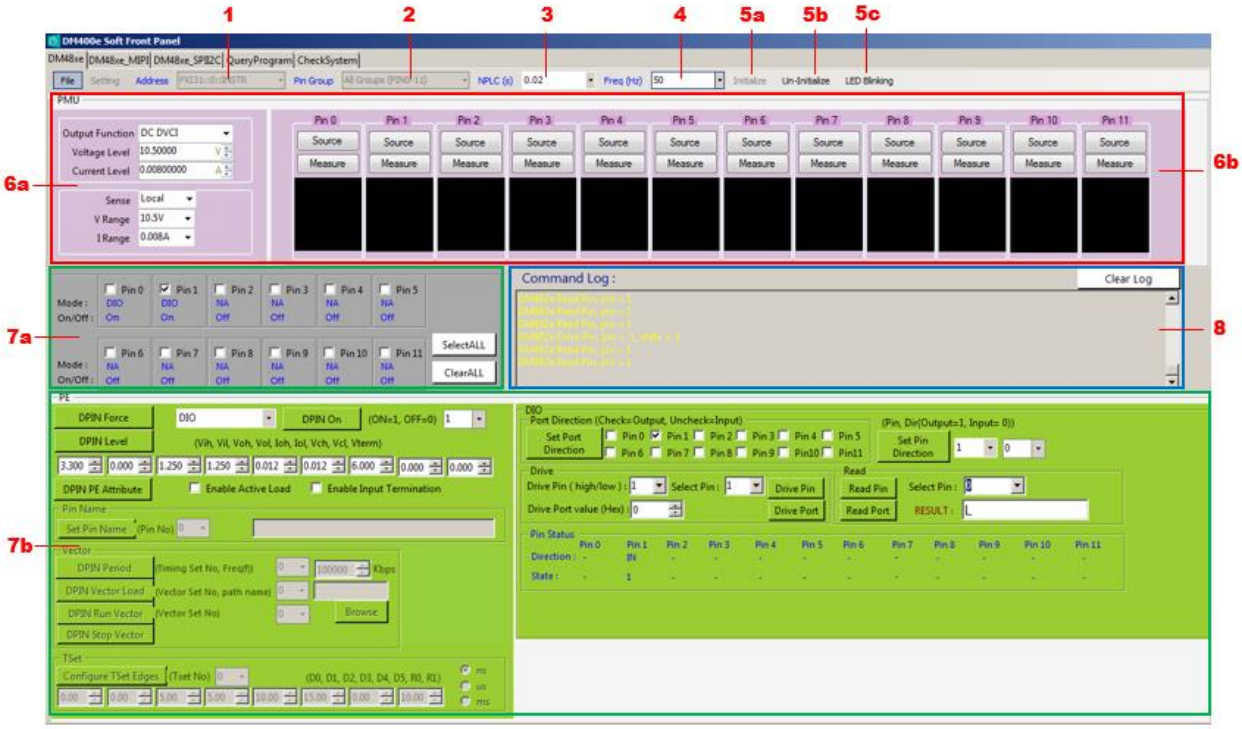



Figure 3: DM481e Soft Front Panel

<table>
<tr><th>Item</th><th>Description</th></tr>
<tr><td>1</td><td>Select the PXI address of the module. A list of addresses is shown for all detected modules in the chassis.</td></tr>
<tr><td>2</td><td>Pin group (The pin group selection will be disabled when using DM481e module).</td></tr>
<tr><td>3</td><td>Set the number of power line cycle, in second.</td></tr>
<tr><td>4</td><td>Set the frequency of the power line (50Hz or 60Hz).</td></tr>
<tr><td>5a</td><td>Initialize the selected module. This includes opening a session to communicate with the module, and then reset the module in its default state</td></tr>
<tr><td>5b</td><td>Un-initialize the selected module. This includes closing the current communication session to the module.</td></tr>
<tr><td>5c</td><td>LED Blinking – Press this button to enable power light blinking at the front panel.</td></tr>
<tr><td>6a</td><td>Per Pin Parametric Measurement Unit (PPMU) function:<br><br>
<table>
<tr><td>Output Function</td><td>DC DVCI: Drive-Voltage Clamp-Current operation. Continuous mode.<br>DC DICV: Drive-Current Clamp-Voltage operation. Continuous mode.</td></tr>
<tr><td>Voltage Level</td><td>Voltage level</td></tr>
<tr><td>V Range</td><td>Range of the voltage level</td></tr>
<tr><td>Current Level</td><td>Current level (Current level setting is not required in DVCI mode. “I Range” will decide the compliance value)</td></tr>
<tr><td>I Range</td><td>Range of the current level</td></tr>
<tr><td>Sense</td><td>Select either Remote or Local sense connection</td></tr>
</table></td></tr>
<tr><td>6b</td><td>Per Pin Parametric Measurement Unit (PPMU) function:<br><br>1. Click the “Source” button for the selected pin to drive the voltage and current level which are defined by user.<br>2. Click on “Measure” button to turn on the SMU and measure the voltage and current level of the selected pin.<br>3. Both buttons will immediately turn to “Green” when the function is ON. Press them once again to turn off the PMU function.<br><br>Note: User may also use it as SMU by turn on “Measure” button only for measurement purpose while the “Source” button is off.<br><br>Following results are shown for each pin:<br>a. Voltage (V)<br>b. Current (A)<br>c. On-board temperature</td></tr>
</table>

7a

Pin Status:

Select/Deselect the individual pin for PE and DIO mode.

| | |
|---|---|
| Mode | Display the configured pin mode in among PMU, PE or DIO mode. |
| On/Off | Shows the status of the pin. |

7b

Pin Electronics (PE):

Generic Setting:

| | |
|---|---|
| DPIN Force | Sets the operation mode:<br><br>Mode: Description<br>VECTOR: Vector engine<br>DIO: General purpose I/O |
| DPIN Level | Set the levels for each pin. |
| DPIN On | Turn on/off the selected pin. |
| DPIN PE Attribute | Enable/disable active load and input termination |

Vector:

| | |
|---|---|
| DPIN Period | Specifies the timing set to be used, as well as its period setting |
| DPIN Vector Load | Load the specified vector set in the vector file to the module |
| DPIN Run Vector | Run the specified vector set. |
| DPIN Stop Vector | Stop the specified vector set. |

TSet:

| | |
|---|---|
| Configure TSet Edges | Configure the Timing Sets Edges. There are total 8 edges setting available (D0, D1, D2, D3, D4, D5, R0 and R1)<br>▪ D0,D3 for direction edges<br>▪ D1,D2,D4,D5 for drive edges<br>▪ R0 and R1 for receive edges |

DIO:

| | |
|---|---|
| Set Pin Direction | Specifies the direction of the selected pin. |
| Set Port Direction | Specifies the direction of the selected pin group. |

<table>
<tr><td></td><td>Drive/Read<br><br><table><tr><td>Select Pin</td><td>Specifies the pin</td></tr><tr><td>Drive Pin Value</td><td>Specifies the logic level of the pin to be driven</td></tr><tr><td>Drive Port Value</td><td>Enter the port value (in HEX) if you want to drive to the whole pin group</td></tr><tr><td>DIO Function</td><td>Click on the available DIO functions:<br>- Drive Pin<br>- Drive Port<br>- Read Pin<br>- Read Port</td></tr></table><br>Click on the DIO function button (Drive Pin, Drive Port, Read Pin, Read Port) for the selected pin to execute the DIO function command.<br>The results for read functions for each pin will be displayed as either “H” (logic high), “L” (logic low) or “Z” (undefined).</td></tr>
<tr><td>8</td><td>Command Log:<br><br>Display the actions taken in the background.</td></tr>
</table>

Table 5: Soft Front Panel Control

Quick steps to configure the module:

1. Select PXIe address of the targeted module from the drop-down list.
2. Press "Initialize".
3. The module is set to drive 10.5V and its current compliance is set to 8mA, in DVCI mode.
4. Press "Source" and follow by "Measure" button to execute the commands. The buttons will turn to green.
5. Both voltage and current are measured.
6. User may configure the selected pin for measurement only by turn off the "Source" button and remain the "Measure" button as "ON".
7. Turn off the pin by click on "Source" and "Measure" buttons to exit from PMU mode.
8. Press "Un-Initialize" to turn off the whole module.

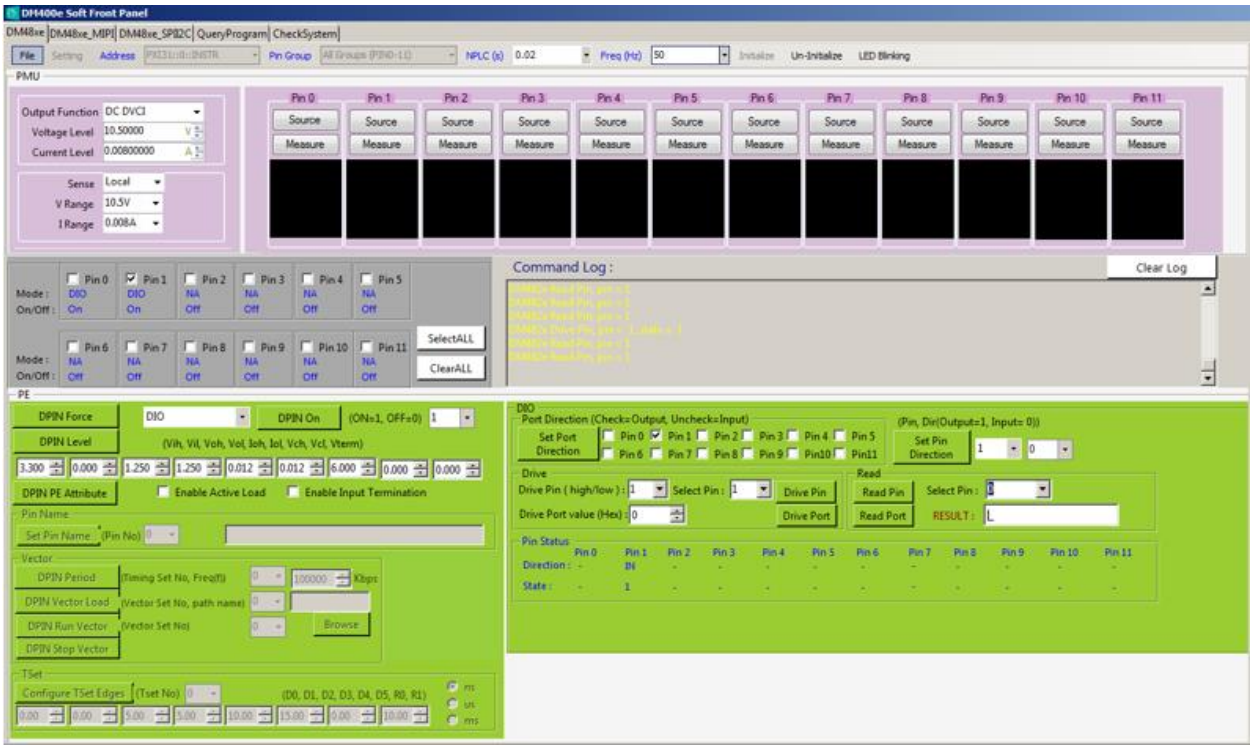



Figure 4: PMU Operation

## 3.3 Remote Operation

Refer to programming manual.

## Section 4: Brief Technical Description

DM481e is a high speed digital waveform generator and analyzer, with integrated pin electronics (PE) and per pin parametric measurement unit (PPMU) functions.

### 4.1 Basic Operation

#### 4.1.1 Per Pin Parametric Measurement Unit (PPMU) Functions

Each DM481e pin can be configured can be set up to function as a voltage source/current monitor, current source/voltage monitor:

1. Source Current, Clamp Voltage (DICV)
    a. This configuration turns DM481e into high-impedance current source
    b. The voltage developed is limited by compliance setting
    c. If current is 0A, DM481e becomes a voltmeter

2. Source Voltage, Clamp Current (DVCI)
    a. This configuration turns DM481e into low-impedance voltage source
    b. Current flow is limited by compliance setting. Current compliance values are limited to the supported current range values. For example, if the available current ranges are 2uA, 8uA, 32uA, 128uA, 512uA, 2mA, 8mA and 32mA, then the available compliance settings are also 2uA, 8uA, 32uA, 128uA, 512uA, 2mA, 8mA and 32mA.

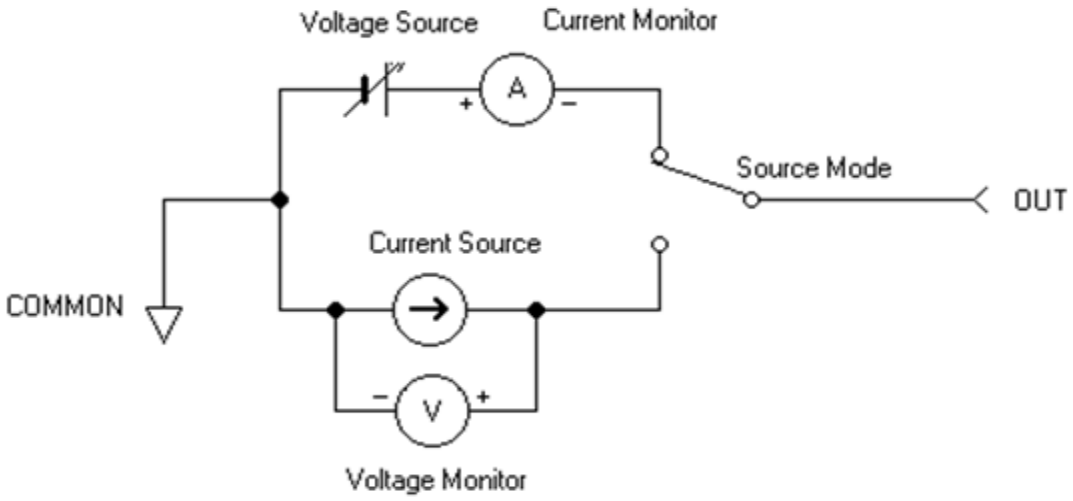



Figure 5: Typical Operation

To prevent over-voltage or over-current damage to the device-under-test (DUT), several levels of output protection, termed COMPLIANCE, is incorporated. Maximum specifiable compliance depends on the voltage or current range at which the pin is operating.
If a source pin reaches compliance during measurement, this indicates an error in the operation.
Example:

1. DM481e is set to voltage source/current monitor mode; compliance of 2mA; output load is a 1kohm resistor.
2. If source voltage is 5V, the current that should flow through the resistor is 5V/1k=5mA.
3. Due to compliance of 2mA, the source channel turns into a constant current source of 2mA. This yields a voltage of 2V across the resistor.

The measurement result is the average value of all samples taken for $n^{th}$ number of power line cycle (PLC). For example, if the line frequency is set to 50Hz (period is 20ms), then 1 PLC equals to 20ms. The measured data is a resultant of averaged of raw samples captured by ADC on DM481e for 20ms.

**NOTE**

The NPLC setting of an instrument allows adjustment of the tradeoff between speed and accuracy. The greater the number of power line cycles, the greater noise rejection and better resolution the signal value will be.

### 4.1.2 Pin Electronics (PE) Functions

Apart from PMU, each DM481e pin can also perform the pin electronic functions of the driver, the comparator, and the active load (DCL), and DC levels for automated test equipment (ATE) applications.

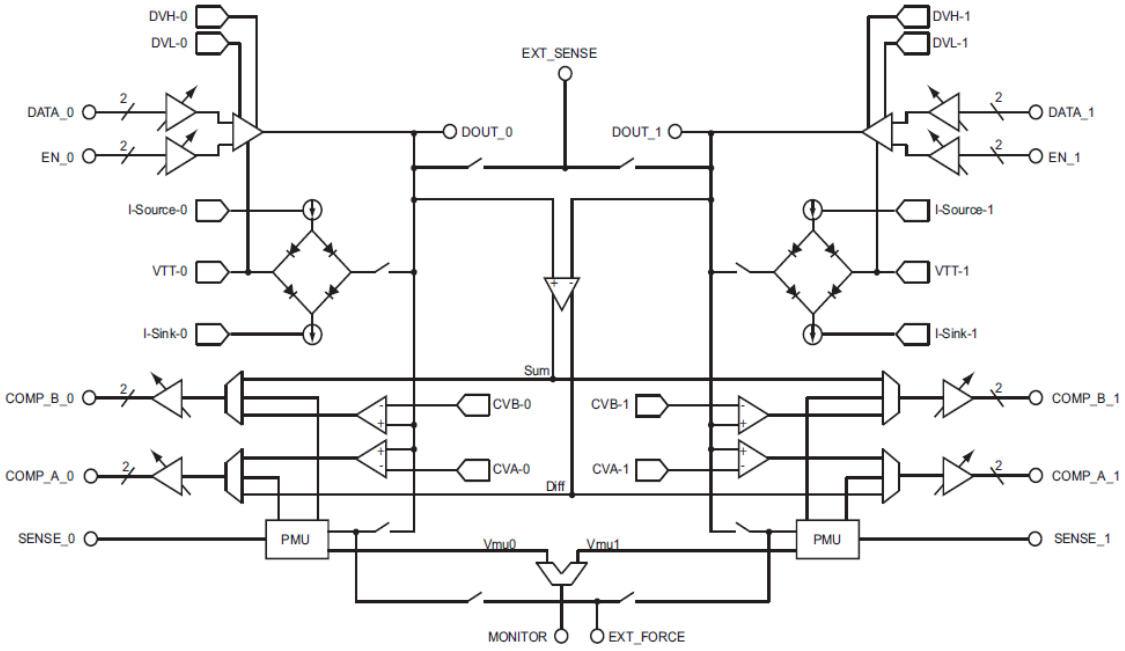



Figure 6: Pin Electronics (PE) Level Setting

Pin level settings:

| Setting | Description |
|---|---|
| DVH | The output driver high voltage level from the pin. |
| DVL | The output driver low voltage level from the pin. |
| CVx | The comparator voltage level into the pin. |
| VTT | The termination voltage when input termination is enabled.<br>If input termination is disabled, driver is in high impedance. |
| I-Sink | The sink current when active load is enabled. |
| I-Source | The source current when active load is enabled. |

Table 6: PE Level Setting

Active load termination is a constant current source. If the external output voltage (e.g. from device-under-test) is more than VT, the pin will sink IOH current, otherwise, the pin will source IOL current.

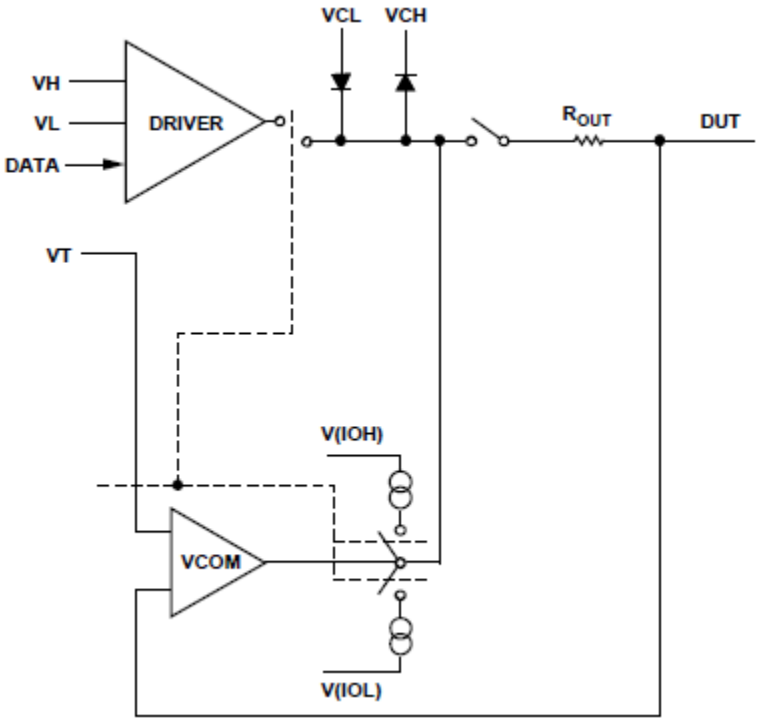



Figure 7: Active Load Block Diagram

### 4.1.3 Pattern

DM481e allows user to generate user-defined patterns and capture incoming waveform for each pin. Two files are required to generate patterns:

1. Declaration file (*.dec). Declaration file consists of user defined Timing Sets, Waveform Sets, pin name declaration and its sequencing
2. Pattern file (*.pat). Pattern files consists of executed Timing Sets, Opcode and Waveform Sets of each channel for each pattern lines. Each pattern files has to be referenced to one declaration file or user may directly instantiate these info (Timing Sets, Waveform Sets and etc) in the pattern file.

DM481e timing formatting feature consists of 8 edges per pin, and is as shown in following figure.

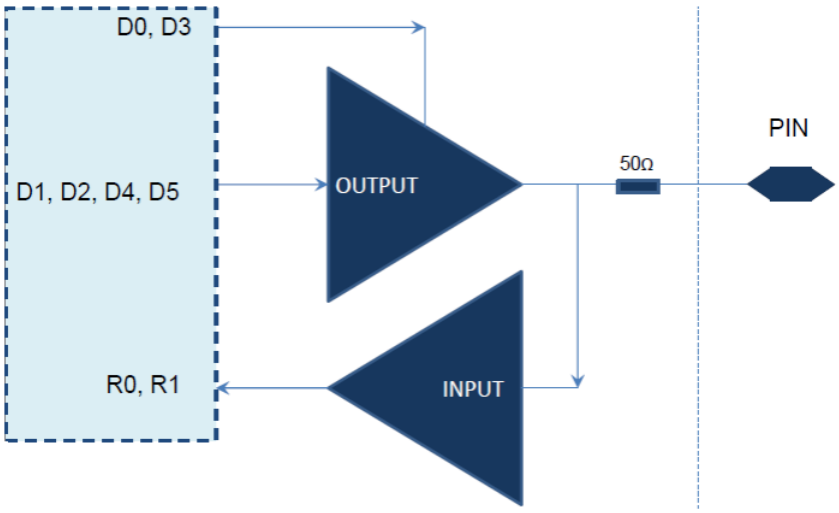



Figure 8: Overview of DM481e Timing Format

| Edge Setting | Description |
|---|---|
| D0, D3 | Directional Edge:<br>It is used to enable or tristate output driver. |
| D1, D2, D4, D5 | Driving Edge:<br>It is used to drive output driver to logic High or Low. |
| R0, R1 | Comparison/Receiving Edge:<br>It is used to compare incoming waveform with respect to logic High, Low, Mid-band or Valid. |

Table 7: Action Setting For 8 Edges

### 4.1.3.1 Timing Sets (TSET)

Timing Sets contains timing information of the 8 edges and waveforms period.

**Rules:**

1. D1 <= D2 <= D4 <= D5
2. D0 <= D3
3. R0 <= R1
4. All the 8 edge values are not allowed to exceed waveform period.

Error will be returned during conversion to binary file if any of the rules above is violated.

Figure 9 and Figure 10 show an example of timing setting that show in declaration file.

User is required to set the pin name in .dec to be in sync with the pin name assigned in amap file through SetPinName API.

```
SEQUENCE
{
    DM_Pin0, DM_Pin1, DM_Pin2, DM_Pin3, DM_Pin4, DM_Pin5, DM_Pin6,
}

TIMINGSET
{
    //Timing set name = period = timing set number;
    TS0 = 40ns = 0;
    //PinName = D0,D1,D2,D3,D4,D5,R0,R1 = Timing set name;
    DM_Pin0 = 6ns,10ns,15ns,33ns,21ns,27ns,2ns,38ns = TS0;
    DM_Pin1 = 0ns,0.00ns,5.00ns,10ns,10ns,15ns,0ns,10ns = TS0;
    DM_Pin2 = 0ns,0.00ns,5.00ns,10ns,10ns,15ns,0ns,10ns = TS0;
    DM_Pin3 = 0ns,0.00ns,5.00ns,10ns,10ns,15ns,0ns,10ns = TS0;
    DM_Pin4 = 0ns,0.00ns,5.00ns,10ns,10ns,15ns,0ns,10ns = TS0;
    DM_Pin5 = 0ns,0.00ns,5.00ns,10ns,10ns,15ns,0ns,10ns = TS0;
    DM_Pin6 = 0ns,0.00ns,5.00ns,10ns,10ns,15ns,0ns,10ns = TS0;
    DM_Pin7 = 0ns,0.00ns,5.00ns,10ns,10ns,15ns,0ns,10ns = TS0;
    DM_Pin8 = 0ns,0.00ns,5.00ns,10ns,10ns,15ns,0ns,10ns = TS0;
    DM_Pin9 = 0ns,0.00ns,5.00ns,10ns,10ns,15ns,0ns,10ns = TS0;
    DM_Pin10 = 0ns,0.00ns,5.00ns,10ns,10ns,15ns,0ns,10ns = TS0;
    DM_Pin11 = 0ns,0.00ns,5.00ns,10ns,10ns,15ns,0ns,10ns = TS0;
```

Figure 9: Timing Settings

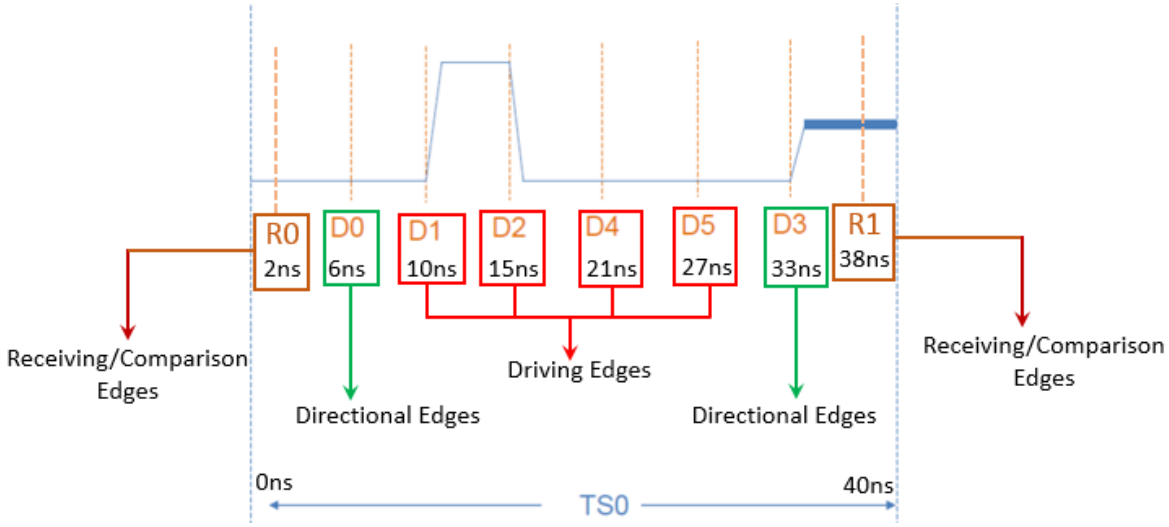



Figure 10: Timing Set Diagram Example

### 4.1.3.2 Waveform Sets (WAVESET)

Waveform Sets contains pattern information for customized patterns and waveforms that are mainly use to configure the 8 edges (D0-D5 and R0 & R1) for each pin to:

- Drive logic High/Low – D1, D2, D4 & D5
- Force On/Off (Enable/Tristate the output driver) – D0 & D3
- Strobe/Window Compare logic High/Low/Mid-band/Valid – R0 & R1

The following figure shows both waveforms are using the same WAVESET pattern but with different TSET.



Figure 11: Waveform with the same WAVESET but different TSET

The following table shows the function of each character that can be used in Waveform Set.

| Character | Action | Description | Applicable Edges |
|---|---|---|---|
| N | No Action | Do nothing | D0-D5, R0, R1 |
| 0 | Drive Low | Drive output data low, when driver is enabled | D1, D2, D4, D5 |
| 1 | Drive High | Drive output data high, when driver is enabled | D1, D2, D4, D5 |
| Z | Force Off | Disable or tristate the output driver | D0, D3 |
| F | Force On | Enable the output driver | D0, D3 |
| L | Compare Low | Edge compare low (VDUT < VOL) | R0, R1 |
| H | Compare High | Edge compare high (VDUT > VOH) | R0, R1 |
| V | Compare Valid | Edge compare valid ((VDUT > VOH) or (VDUT < VOL)) | R0, R1 |
| M | Compare Mid-band | Edge compare mid-band (VOL < VDUT < VOH) | R0, R1 |
| l | Compare Low Window | Window compare low (VDUT < VOL) | R0 |
| h | Compare High Window | Window compare high (VDUT > VOH) | R0 |
| v | Compare Valid Window | Window compare valid ((VDUT > VOH) or (VDUT < VOL)) | R0 |
| m | Compare Mid-band Window | Window compare mid-band (VOL < VDUT < VOH) | R0 |
| c | Close Window Compare | End of window compare | R1 |

Table 8: Symbol and Action for WAVESET

Note:

1. All the symbols that used in Waveform Set are case sensitive.
2. If any "h", "l", "m" or "v" is used on R0 for window compare, then "c" must be placed on R1 in order to close the window compare.

The following figure shows an example of Waveform Set setting.

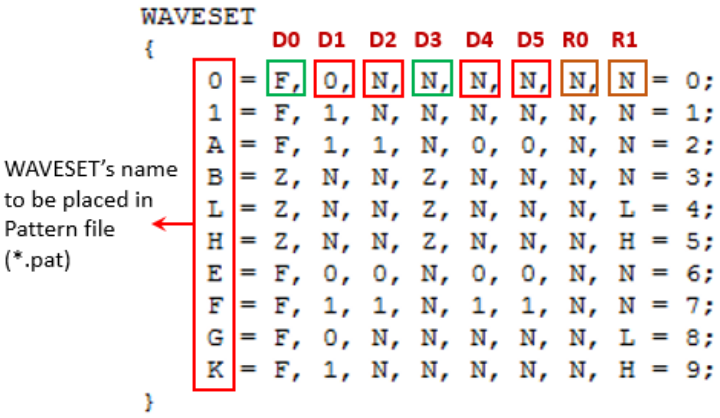



Figure 12: Waveform Set Settings

Some examples of Waveform Sets as shown below:

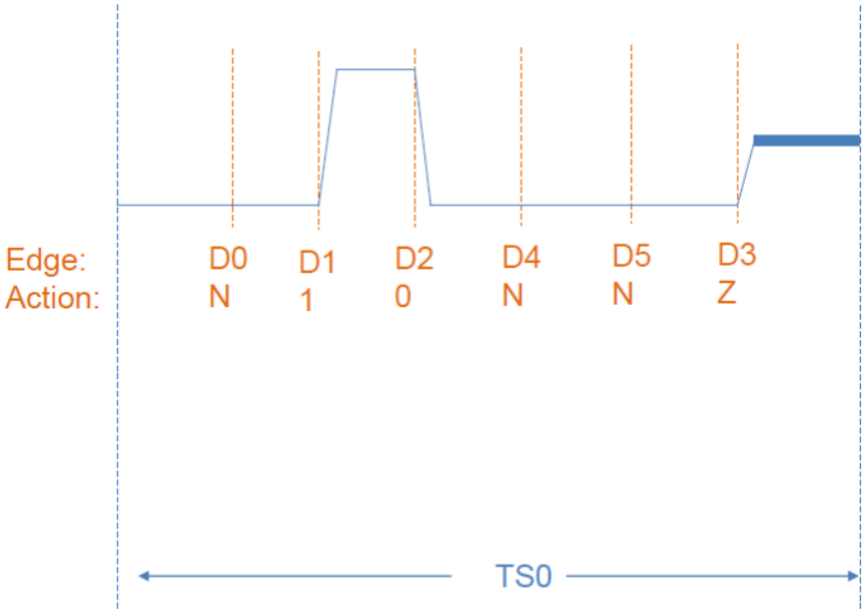



Figure 13: Example 1 of WAVESET

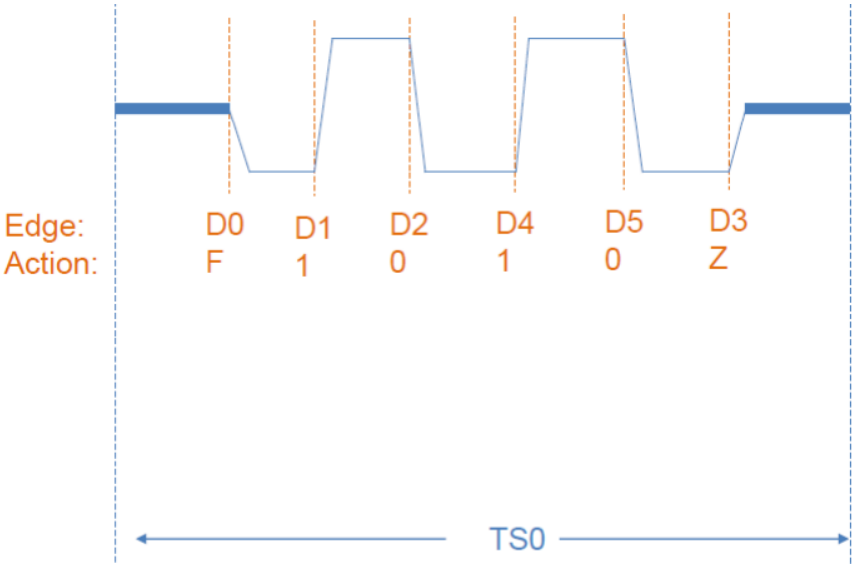



Figure 14: Example 2 of WAVESET

There are 2 types of logic comparison options available in DM481e:

1. Strobe Comparison
   A strobe signal compares the logic of the signal at one specific time. The figure below shows an example of strobe comparison.

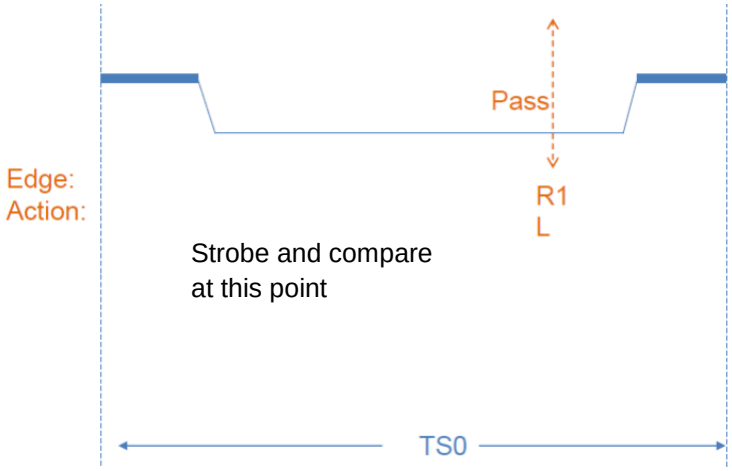



Figure 15: Strobe and Compare Data

2. Window comparison
   Window comparison performs multiple comparison at the resolution of 1.25ns within the window frame. Any single comparison failure within this window frame is flagged as fail to the user. The figures below show examples of window comparison.

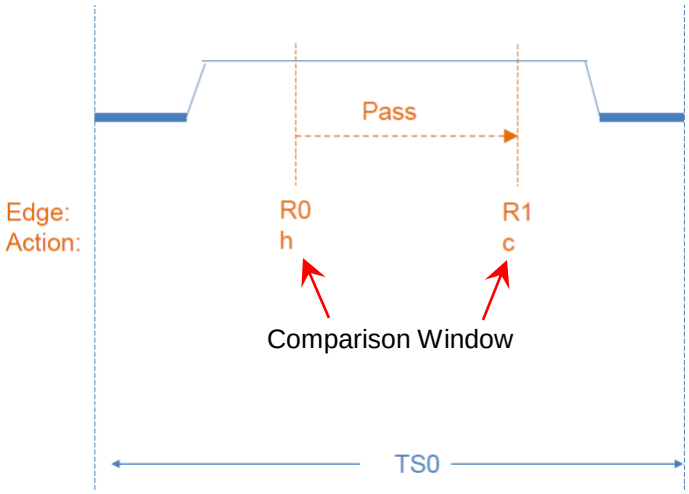



Figure 16: Window Comparison Resulted in Pass

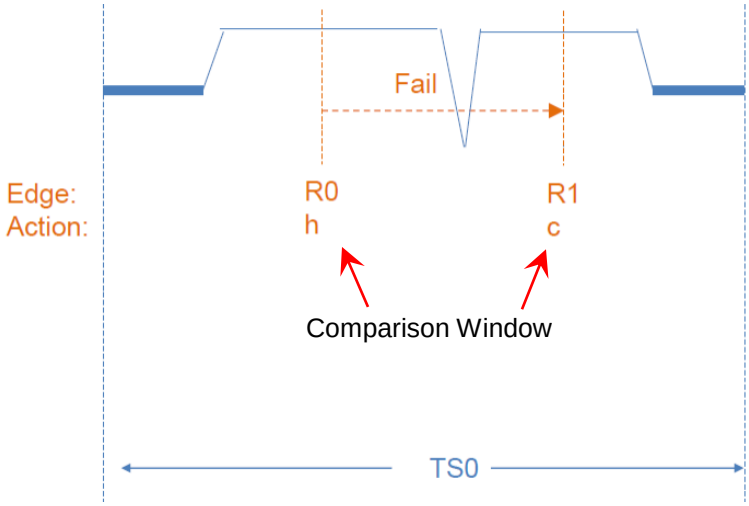



Figure 17: Window Comparison Resulted in Fail

### 4.1.3.3 Opcode & Operand

DM481e supports opcode to perform some actions along with the execution of a pattern line. Operand acts as an argument to certain opcode to perform this opcode in different behavior. For example, the operand of opcode "rep" allows user to repeat that pattern line for different number of times.

Below are the supported Opcode to be used in Pattern Generation:

| Opcode | Operand | Description |
|---|---|---|
| nop | - | No operation. |
| str | - | Store into History RAM (HRAM). |
| trig | 8-bits output trigger. | Drive the trigger outputs. |
| rep | x | Repeat current vector for x times. |
| stop | - | End of pattern. |

Table 9: Opcode & Operand

Note:
Every pattern files requires opcode stop to be used at the last pattern line.

#### 4.1.3.4 Pattern (*.pat)

Pattern files consists of executed Timing Sets, Opcode and Waveform Sets of each channel for each pattern lines.

Note:
Please refer to DM481e Programming Manual and Specifications for more details about the API application.

The figure below shows an example of a pattern file.

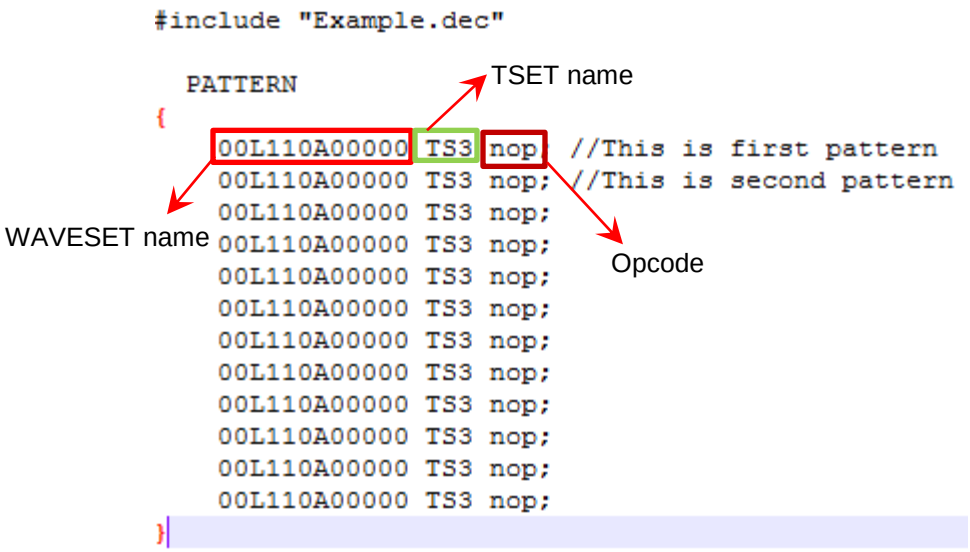



Figure 18: Pattern File

#### 4.1.3.5 Pattern Compiler (*.exe)

Once a pattern file is done, it is required to convert pattern file to a binary equivalence (*.pbin) through a pattern compiler – "tvc2.dm481e.exe".

User is required to prompt to the directory that stores all the "tvc2.dm481e.exe", "*.pat file" and "*.dec file" and execute the following command in a Command Prompt window:

- tvc2.dm481e.exe example.pat

As an output, the example.pbin file will be generated.

User has to load this generated binary file into DM481e by calling the DPINVecLoad API.

### 4.1.4 Triggers

Figure below shows the triggering signals supported by DM481e.

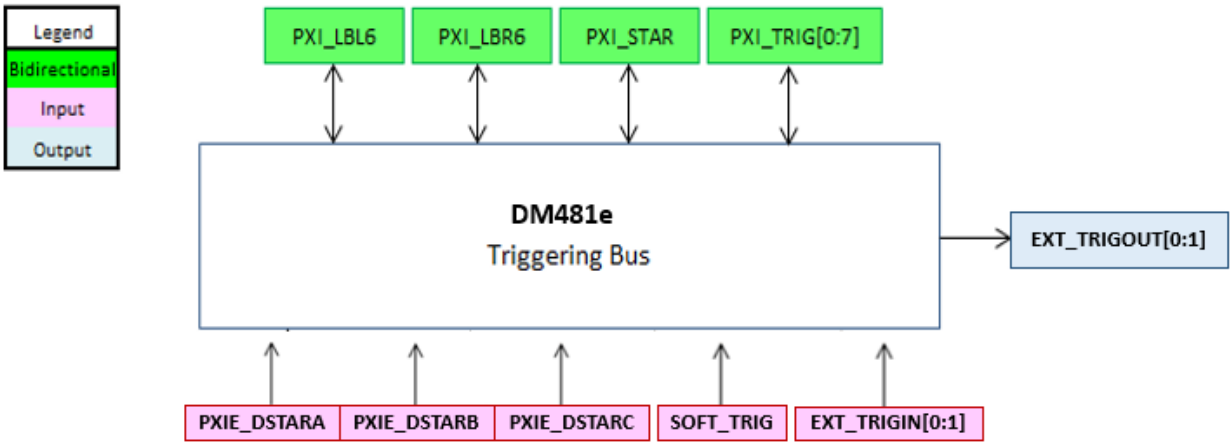



Figure 19: DM481e Trigger Bus

## 4.2 Signal Connection

There are 4 leads per pin:

1. Force-High (FH)
2. Sense-High (SH)
3. Low (L)
4. Common Sense-Low (SL)

Force-High and Low signals are where voltage and current are generated during source or sink operation. Sense-High and Common Sense-Low signals are high impedance paths. Connect these lines as close to the DUT terminal as possible for more accurate voltage source or measure operations. Presence of Sense lines at the DUT terminals is known as 4-wire, remote-sense, or Kelvin operation.

## Section 5: Maintenance

### 5.1 Cleaning

Before doing any cleaning to the product, switch off the chassis and disconnect it from main power supply.

You can wipe the front panel of the module using a soft cloth moistened in water, taking care not to wet the connectors. Do not use aerosol or liquid solvent cleaners.

You can use a handheld vacuum cleaner to remove any accumulated dust on the product, if necessary.

**CAUTION**

Be careful NOT to be in direct contact with any circuits on the product to prevent any damages. Contaminants such as oils and salts from the human skin can affect the performance of the product.

Comply with good ESD practice when handling the product during cleaning.

### 5.2 Calibration

Electronic components tend to drift over time. Environmental changes, operation stresses and aging increase the measurement uncertainty. Periodic calibration is required to ensure the accuracy of the instrument.

The recommended calibration interval for the product is 12 months.

### 5.3 Servicing

There are no user-serviceable parts in the product; if any attention is needed, return it to instrument manufacturer.

**NOTE**

Any repair that is not covered in this manual should only be performed by qualified Aemulus personnel.

## Section 6: Warranty

Aemulus warrants that the product delivered will be free from defects in material and workmanship for 1 year from the date of delivery order. This warranty does not cover the product if it is damaged in the process of being installed.

This warranty does not cover replacement of products damaged by abuse, accident, misuse, neglect, alteration, repair, improper installation or improper testing. If the product is found to be defective otherwise, Aemulus, at its option, will replace or repair the product at no charge. If you ship the product, you must assume the risk of damage or loss in transit. Aemulus may replace or repair the product with either a new or reconditioned unit, and the product returned becomes Aemulus property.

## Section 7: Revision History

| | | |
|---|---|---|
| 2.0 | JUNE 2017 | INITIAL RELEASE (FOR REV3 DM481E) |
| 2.1 | JULY 2017 | UPDATED DESCRIPTION IN SECTION 4.1.4 TRIGGERS |

## Section 8: Contact Us

To obtain service, warranty or technical assistance, please contact Aemulus.

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Product specifications and descriptions in this document are subject to change without prior notice.